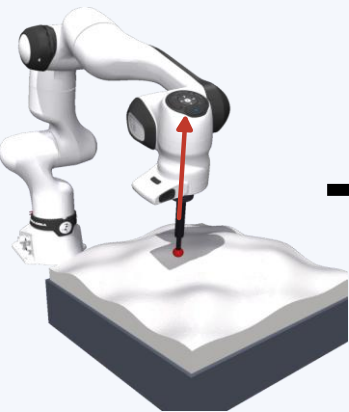


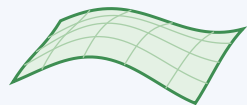
Input

State $(q, \dot{q}) \in \bar{\mathcal{M}}$



Constraint

Contact Manifold



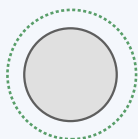
$$c(q) = 0, J_c \dot{q} = 0$$

Dynamic Wrench Inequality



$$\mathcal{K}_i = \{\lambda : A_i \lambda \leq b_i\}$$

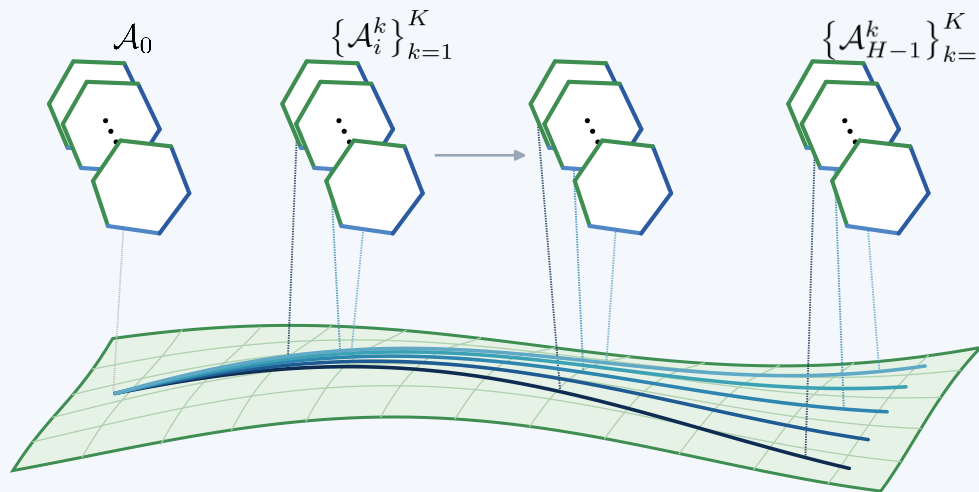
Kinematic Inequality



$$h_j(q) \geq 0$$

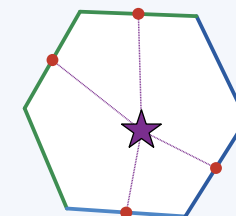
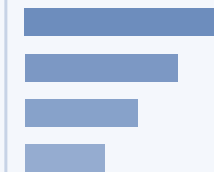
One control step – Constrained rollout

Admissible set rebuilt at every rollout state $\mathcal{A}_t^k = \mathcal{A}(q_t^k, \dot{q}_t^k)$



Output

$$w_k \propto e^{-J_k/\beta}$$

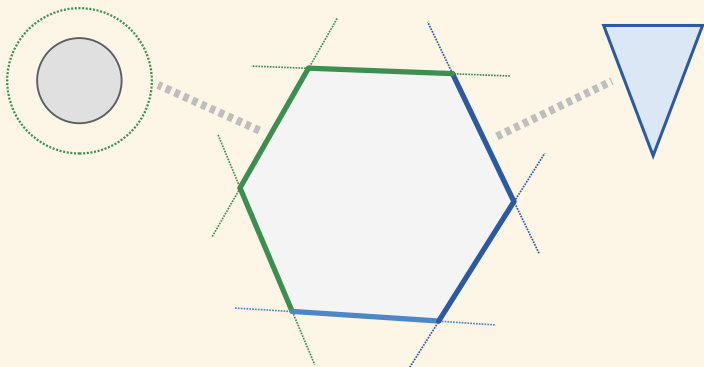


$$a_0^{\text{exec}} = \sum_k w_k a_0^{k,*} \in \mathcal{A}_0$$

Inside a single rollout step

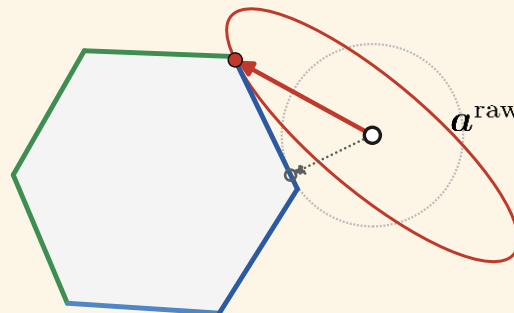
1) Compose one admissible set

$$Ca \leq d \iff C^\lambda, C^h \text{ stacked}$$



2) Project under the inertia metric

$$a^* = \arg \min \frac{1}{2} \|a - a^{\text{raw}}\|_M^2$$



3) Carry the set onto the manifold

$$F\mathcal{A} + f \subset \{\ddot{q} \mid J_c \ddot{q} + \dot{J}_c \dot{q} = 0\}$$

